

DEEP DIVE · CASE 023

The slab did not have tomorrow's strength today

Skyline Plaza · Bailey's Crossroads, Virginia · 2 March 1973 · flat-plate construction

At about 2:30 p.m. on 2 March 1973, part of Skyline Plaza building A-4 collapsed through the height of the partially constructed tower and into the adjoining parking garage. Fourteen construction workers were killed and 34 were injured. The National Bureau of Standards concluded that premature removal of supporting forms was a contributing factor and identified the most likely initiating mode as shear failure around one or more columns in the 23rd-floor slab.

THE BUILDING HAD SEVERAL CLOCKS RUNNING AT ONCE

A typical floor was poured in four sections. NBS reconstructed casting dates near the top of the building: 23rd-floor sections were poured on 23, 24, 26 and 27 February, while portions of the 24th floor were poured on 28 February, 1 March and the morning of 2 March. At collapse, parts of the 23rd floor were only about three to seven calendar days old.

NBS estimated average in-situ strength for the 23rd-floor slab at about **1200 psi** for the four-day condition used in one analysis, compared with the **3000 psi** specified 28-day design strength. Construction scheduling had converted time into a structural variable.

YOU ARE THE ENGINEER

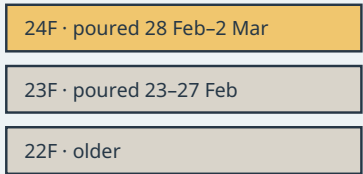
A contractor asks to strip shores because the programme says the next floor must start. The mix is specified at 30 MPa at 28 days. What evidence lets you say yes?

Not the 28-day mix designation. You need evidence tied to the actual state: field-cured or maturity-correlated strength; loads already present; loads about to arrive; the shoring or reshoring arrangement; and the resulting shear, flexural and deflection demand.

NBS CASE 1 MAKES THE HIDDEN LOAD VISIBLE

One NBS analysis assumed the 23rd-floor slab was unshored. Under the portion of the 24th floor just cast, the 23rd-floor system carried about **80 psf** from the new slab, **5 psf** from formwork and **80 psf** from its own weight: **165 psf** before future occupancy entered the conversation.

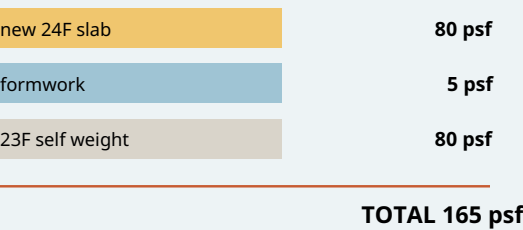
CONSTRUCTION AGE STACK AT COLLAPSE



Different parts of one floor had different ages.

Construction age was not a single value across the floor.

NBS CASE 1 — LOAD ON 23RD FLOOR



The report found the 23rd-floor slab more critical in shear than in flexure. Once one column support was lost, the remaining supports could be overstressed; debris from upper slabs then overloaded lower slabs and drove progressive collapse downward.

THE SHORING MAP IS PART OF THE STRUCTURAL MODEL

NBS could not establish every shore and reshore position with certainty. Employee statements conflicted, and additional shoring was installed after the collapse before investigators gained full access. If safety depends strongly on temporary support, document it well enough that the engineer can state what structure exists at each stage.

ENGINEER'S NOTEBOOK · NOTE-006

Capacity is conditional on state, continuity and interaction. Ask which age, moisture state, support arrangement, load path or system interaction makes the quoted capacity available.

SOURCES FOR THIS CASE

Primary · Tier A — NBS/NIST, Investigation of the Skyline Plaza Collapse in Fairfax County, Virginia (NBS BSS 94), 1977.
[NIST publication](#)

SITE PROBLEM · CASE 024

Fill is a construction product, not just a soil name

Sau Mau Ping · Hong Kong · 1976 disaster / GEO review

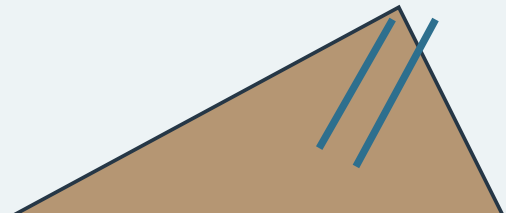
The 1976 Sau Mau Ping fill-slope failure killed 18 people after wet mud entered the lower floors of Block 9. The independent review panel concluded that water penetrated the slope face, created a wetted zone and reduced the strength of loose fill until downhill movement developed and rapidly converted into a mud avalanche. Investigations showed that the slope face had been formed by **end-tipping fill in a loose condition**.

That makes this an earthworks-quality problem as much as a rainfall problem. Two slopes can have identical geometry and the same broad soil name while behaving very differently because one has been placed and compacted as engineered fill and the other has been tipped loose down a face.

YOU ARE THE ENGINEER

You inherit an old fill slope behind a building. The drawings say “compacted fill,” but there are no placement records. Treat the label as a hypothesis: investigate density, layering, services, seepage and drainage rather than assuming the specification describes the workmanship.

LOOSE END-TIPPED FILL + WETTING



Placement history changes the available shear strength.

ENGINEER'S NOTEBOOK

Ask how the fill got there. Placement history is a material property in disguise.

SOURCES FOR THIS CASE

Primary · Tier A — Hong Kong GEO, Report of the Independent Review Panel on Fill Slopes, GEO Report No. 86.
GEO report

DETAIL / ROBUSTNESS · CASE 025

The wall carried load; the system could not lose the wall

Ronan Point · London · 16 May 1968 · large-panel precast construction

1 NORMAL



2 WALL LOSS



3 CASCADE



Local support loss became a system problem because there was no robust alternate load path.

A domestic gas explosion on the 18th floor of Ronan Point blew out load-bearing wall panels. The official inquiry found that the explosion caused the local wall loss and that the ensuing partial collapse was inherent in the building's design. NIST later described a precast wall-and-floor system whose floors were supported directly by wall panels and whose connections offered no redundant load path after a local bearing wall disappeared.

The upper floors first lost support and fell. Their impact then exceeded the capacity of lower floors and drove collapse downward. The central diagnosis is **lack of structural integrity**: no alternate path existed at the onset of bearing-wall loss.

Connection design therefore cannot stop at “can the panel carry its normal reaction?” Robustness asks whether floors can tie together, span differently or otherwise bridge a local loss long enough to prevent disproportionate collapse.

DETAIL CHECKLIST

Bearing + tying + alternate path + debris consequence. The first proves the normal state; the others address the damaged state.

SOURCES FOR THIS CASE

Primary · Tier A — UK National Archives, Ronan Point inquiry material, 1968. [National Archives](#)

Supporting · Tier A — NISTIR 7396, Best Practices for Reducing the Potential for Progressive Collapse in Buildings.

CIVIL ENGINEERING WIN · CASE 026

The line of defense became a system

Greater New Orleans HSDRRS · post-Katrina reconstruction

The Hurricane and Storm Damage Risk Reduction System around greater New Orleans is useful precisely because its achievement cannot be reduced to one stronger levee. FEMA accredited the system in 2014 after determining that the improvements reduce risk from effects associated with the 1-percent annual chance storm. USACE reported that nearly **133 miles** of levees, floodwalls, gated structures and pump stations had been strengthened or improved.

New surge barriers and closure complexes moved the primary line of defense outward, removing about **68 miles** of interior levees and floodwalls from direct storm-surge exposure.

CLOSING THE HYDRAULIC BOUNDARY

A flood-defense system has an awkward property: the weakest open path can govern the result. A taller levee does little if an adjacent gate cannot close. A closed perimeter creates a new problem if rainfall and pumped drainage cannot get out. A structurally strong crest can still lose the embankment if overtopping erodes the protected-side slope.

HSDRRS addresses those conditions with a portfolio rather than a single section type: perimeter levees and floodwalls, surge barriers, gates, pump stations and armoring.

SYSTEM REVIEW — FOUR QUESTIONS

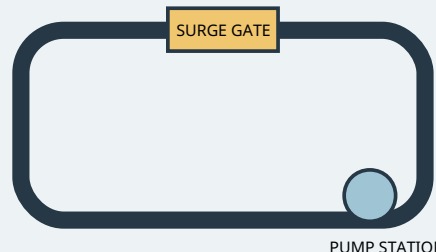
Where is the boundary? Define the full line that must resist water.

How does it close? Identify gates, penetrations and transitions.

What happens inside? Account for rainfall, seepage and pumping.

How does capacity change with time? Settlement, erosion, corrosion and mechanical reliability alter the future state.

THE CAPACITY IS IN THE CLOSED PERIMETER



levees + floodwalls + gates + pumps + armoring must work in one event

The most interesting design move was to change **where** surge is resisted. USACE describes surge barriers at Lake Borgne, Seabrook, the outfall canals and the West Closure Complex as pushing the line of defense outside the city. That reduces the length of interior defenses directly exposed to surge, changing the topology of the risk problem rather than merely thickening every existing component.

A SYSTEM CAN BE ACCREDITED AND STILL NEED FUTURE WORK

FEMA accreditation is not a certificate of flood-proof immortality. USACE continues work on armoring, pumping and future levee lifts because consolidation, settlement, subsidence and sea-level conditions change the available elevation over time. A crest elevation is a depreciating asset unless the system is monitored and maintained.

For ordinary civil works, draw the full system boundary and mark every closure, low point, outlet, transition and maintenance-dependent component. Strong components do not average out an open path.

ENGINEER'S NOTEBOOK

Review continuity before average strength.

A protection system succeeds when every required element is available in the same event.

SOURCES FOR THIS CASE

Primary · Tier A — USACE / FEMA, FEMA Accredits Hurricane and Storm Damage Risk Reduction System (HSDRRS), 2014. USACE release

GEOTECHNICAL ENGINEERING WIN · CASE 027

Test the foundation system you actually designed

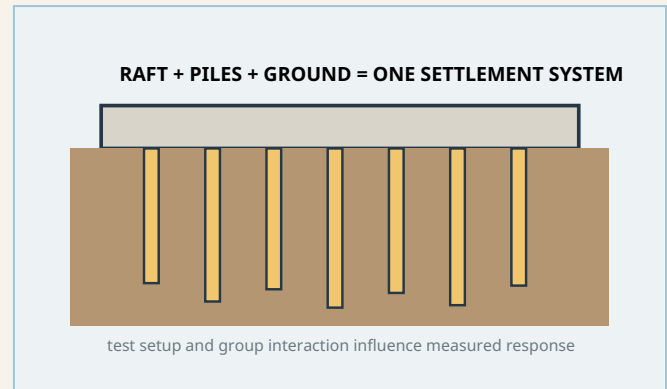
Burj Khalifa · Dubai · piled raft foundation

The Burj Khalifa foundation was designed as a **piled raft** on deep carbonate soils and rocks. The design programme combined ground investigation, numerical analysis, peer review and full-scale pile testing. The 2008 project paper reports trial piles loaded to as much as **64 MN**, with predicted load-settlement behaviour compared against measured response and tower settlements monitored during construction.

A piled raft is a useful antidote to component thinking. The piles do not act in splendid isolation while the raft politely watches. Raft contact, pile stiffness, pile-soil interaction, ground stiffness and the superstructure all influence the settlement field.

A later peer-reviewed re-assessment used two piled-raft analysis methods and found calculated average and differential settlements in reasonable agreement with measured data near the end of construction. It also emphasised that reaction piles used in a load test can influence the interpreted test-pile response.

That last point is wonderfully transferable: even the **test** has a system boundary. A number read from an instrument is not self-interpreting evidence if the test setup changes the response being measured.



PRACTICAL TRANSFER

For an ordinary piled building, ask whether the design is being treated as “pile capacity × number of piles” when settlement, raft contact or group interaction actually governs. Where pile tests are used, document how the setup, reaction system, loading sequence and interpretation map back to the foundation model.

The evidence is strong for documented design, testing and performance, but the main design paper is authored by project specialists. It is not an independent forensic audit and does not make piled rafts universally preferable.

SOURCES FOR THIS CASE

Primary · Tier B — Poulos & Bunce, Foundation Design for the Burj Dubai - the World's Tallest Building, 2008. [Project paper](#)
Supporting · Tier A — Russo et al., Re-assessment of foundation settlements for the Burj Khalifa, Dubai, Acta Geotechnica, 2013. [Journal paper](#)

THE THREAD · CAPACITY IS CONDITIONAL

Skyline Plaza says concrete strength belongs to a date and support arrangement. Sau Mau Ping says fill strength belongs to placement and wetting history. Ronan Point says local bearing capacity is not the same as system robustness. HSDRRS says flood-defense capacity belongs to a closed and maintained perimeter. Burj Khalifa says foundation performance belongs to interaction between raft, piles, ground, test setup and structure.

The recurring engineering move is to replace “**what is the capacity?**” with “**what conditions make that capacity available?**” The second question exposes time, water, continuity, construction sequence and interaction — exactly the things a tidy calculation is most tempted to hide.

60-SECOND TAKEAWAY

- Never borrow 28-day concrete strength for a younger construction stage without evidence.
- Treat placement and compaction history as part of the material definition for fill.
- For critical supports, check the damaged state and alternate load path, not only normal bearing.
- Review flood and drainage defenses as continuous systems with closures, outlets and maintenance-dependent elements.
- Use full-scale tests and monitoring to calibrate the foundation system, not merely to decorate the calculation report with data.

ARCHIVE RECALL

Which earlier Casebook case showed that a temporary support system becomes the primary gravity system while the permanent load path is unavailable?